MetaWingman: a question-first, step-verified, self-improving agent for systematic review and meta-analysis

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Abstract

Background. Systematic reviews and meta-analyses (SR/MAs) remain labor-intensive and error-prone despite mature reporting standards. Recent AI research agents automate pieces of the pipeline, but few bind the review to a derived question, verify work at the level of individual methodological steps, and leave an auditable record through which the system improves itself.

Methods. We built MetaWingman, a five-component agent: a Review Question Certificate that derives a falsifiable question through a seven-stage process with hard and soft gates; a ten-stage Socratic checklist that gates stage entry; a step-level verifier for appraisal dossiers with abstention and a human-review window; a JSONL audit log and meta-update loop through which source-carrying proposals are applied after review; and retrieval components for section-role classification and evidence retrieval. Components were trained on 109,028 deterministically weak-labeled examples. Validation proceeds through a six-level ladder (VAL-1, 2a, 2b1, 2b2, 2c, 3) ending in time-sliced reconstruction of published reviews.

Results. Section-role classification reached macro-F1 0.9995 and evidence retrieval MRR 0.962 (P@1 0.933). Open-set retrieval favored BM25 single-stage (MRR 0.2649) over trained rerankers, which hurt two-stage performance. A deterministic R pipeline reproduced a published random-effects meta-analysis (pooled MD 2.865 vs 2.9; I^2 92.67% vs 93%) in three locked runs. A first AI-only screening pilot on 649 frozen records achieved gold recall 0.765 (114/149), with abstention the dominant loss. Cross-provider agreement reached kappa 0.8472.

Conclusions. MetaWingman demonstrates a question-first, step-verified architecture with reproducible reconstruction evidence. Current component metrics support consistency with weak labels, not scientific validity; an AI-only end-to-end review remains unfinished.

Keywords: systematic review; meta-analysis; AI agent; question certificate; step-level verification; living systematic review

中文摘要

背景。系统综述与 meta 分析虽有成熟的报告标准，仍耗时且易错。近年的科研 AI 智能体自动化了流程的若干环节，但很少有系统把综述绑定到一个经过推导的问题上、在方法学步骤层面做验证，并留下可审计记录使系统自我改进。

方法。我们构建了 MetaWingman，一个五组件智能体：七阶段推导且带硬/软门的综述问题证书；十个阶段的苏格拉底清单门禁；带弃权与人工审核窗口的评价步骤级验证器；带出处的 JSONL 审计日志与元更新回路；以及 section-role 分类与证据检索组件。组件在 109,028 条确定性弱标签样本上训练；验证经由六级阶梯（VAL-1 至 VAL-3），终点是已发表综述的时间切分重建。

结果。 section-role 分类 macro-F1 0.9995，证据检索候选集 MRR 0.962（P@1 0.933）。开集检索以 BM25 单阶段（MRR 0.2649）为优，训练重排器在两段式配置下为负贡献。确定性 R 管线以三次锁定运行复现了已发表的随机效应 meta 分析（合并 MD 2.865 对 2.9；I^2 92.67% 对 93%）。首个 AI-only 筛选 pilot（649 条冻结记录）黄金召回 0.765，弃权为主要损失；跨提供方一致性 kappa 0.8472。

结论。 MetaWingman 展示了提问先行、步骤验证、可审计自改进的架构，并附复现实证。当前组件指标支持的是与弱标签/准则标签的一致性，而非科学有效性；AI-only 的端到端综述仍未完成。

关键词：系统综述；meta 分析；AI 智能体；问题证书；步骤级验证；living systematic review

1. Introduction

Systematic review and meta-analysis is the backbone of evidence-based medicine, yet it is slow, expensive, and sensitive to process errors at every stage, from search construction to risk-of-bias judgment [1]. The conduct standards are mature: the Cochrane Handbook for Systematic Reviews of Interventions defines the methods [1], PRISMA 2020 governs reporting [2], PRISMA-S extends reporting to the search [3], and living systematic review frameworks specify how a review should be updated rather than frozen [4]. What remains unresolved is execution. Each of the ten stages a review passes through, topic, protocol, search, screening, extraction, appraisal, analysis, writing, reproducibility, and update, is a place where undocumented shortcuts enter, and where an AI assistant can either compound or contain the damage.

The practical burden is not just volume. A review is a chain of judgments in which every downstream decision inherits the assumptions of the ones before it. A search that misses a database, a screening rule that silently excludes a population, an extraction that transcribes the wrong arm, and an appraisal that over-weights a domain all move the final estimate, and none of them is visible in the published forest plot. The standards documents tell a review team what must be done; they do not detect when a step was skipped. This asymmetry, between well-specified methods and weakly monitored execution, is the specific problem an agent for systematic review must solve, and it shapes the design choices in this paper.

The past two years have produced a wave of research agents. The AI Scientist drafts and reviews machine-learning papers end to end [5], AI Scientist-v2 adds a multi-module architecture with self-correction [6], and Google' s AI co-scientist frames discovery as a generate-debate-evolve loop over hypotheses [7]. Agent Laboratory targets research workflows including systematic review [8], PRefLexOR introduces recursive preference optimization for scientific reasoning [9], and Reflexion [10], process verifiers [11], and later reflection architectures [12] established the general pattern of an agent that inspects its own outputs. MetaSyn explicitly targets systematic review and meta-analysis as an agentic pipeline [13], and FirstResearch contributes a question-certificate mechanism that derives and gates review questions before work begins [14].

What these systems share is a gap at the level of the review' s scientific contract. First, the question is usually treated as input, not as a derived, gated artifact; a review built on an underexamined question cannot be rescued by better execution. Second, verification, where present, operates on finished outputs (a manuscript, a hypothesis list), not on individual methodological steps such as an appraisal judgment; a wrong intermediate step can propagate silently into a correct-looking synthesis. Third, self-improvement claims are rarely backed by an auditable trail that ties a proposed change to the failure that motivated it. Fourth, and most consequential for the methods literature, capability claims rest on demos rather than on reconstruction: re-running a published review from its own data and comparing against the published numbers, which is the closest thing to a ground truth that the field has.

We report MetaWingman, an agent built on five components that address these gaps directly: a Review Question Certificate migrated from FirstResearch [14]; a ten-stage Socratic checklist that gates stage entry; a step-level verifier for appraisal dossiers; a JSONL audit log with a meta-update loop; and retrieval components for section-role classification and evidence retrieval. We describe the training pipeline built on deterministically weak-labeled data, a six-level validation ladder, and time-sliced reconstruction of published reviews. We report measured results from execution receipts: component metrics, an open-set retrieval comparison that argues against trained rerankers, a locked three-run reproduction of a published meta-analysis, a first AI-only screening pilot, and cross-provider agreement. We do not claim validated equivalence with human review; the evidence we present supports consistency with rubric-grounded labels and deterministic reproduction within a defined engine boundary.

2. Methods

2.1 System overview

MetaWingman is organized as five components whose outputs are written to a project control plane (review state, protocol, event ledger) and validated before they may alter scientific artifacts. The five components are (1) the Review Question Certificate (RQC), (2) the ten-stage Socratic checklist, (3) the step-level verifier, (4) the audit log and meta-update loop, and (5) the retrieval components. Table 1 summarizes each component, the mechanism it implements, and the evidence reported in this paper.

The control plane separates three kinds of state that are easy to conflate in an agentic pipeline: the review's protocol and derived artifacts (what the review claims to do), the event ledger (what the system actually did, with input and output hashes), and the human responsibility record (which decisions a person signed). A component's output enters the review only after the relevant validator accepts it; a failed validator produces a blocked or abstained event, never silent continuation. This separation is what makes the reconstruction experiments in Section 3 possible: because every run is logged with hashes and frozen configuration, a reproduction claim can be checked against the log rather than trusted from a narrative.

2.2 Review Question Certificate

The RQC is migrated from FirstResearch [14] and implements the question-first principle: the review question is derived, not assumed. The generator proceeds through seven stages that move from primitives to a falsifiable hypothesis: (i) extraction of primitives (population, intervention/exposure, comparator, outcome, setting), (ii) statement of first-principle assumptions, (iii) construction of a mechanism model linking the question to the evidence landscape, (iv) articulation of the underlying tension the review will resolve, (v) derivation of a falsifiable hypothesis, (vi) specification of a minimal decisive test, and (vii) a novelty gate. The novelty gate is a hard gate backed by a Europe PMC novelty search plus a soft gate for answerability and decision relevance; the certificate schema carries both gate types and a failure-update rule stating what evidence would force revision of the question. Generation uses max_tokens 8192 (evidence: `metawingman/scripts/generate_review_question_certificate.py`). To assess the certificate independently of the generator's self-report, we ran a double-judge blind evaluation in the LLM-as-a-judge paradigm [15]: two judge models scored certificates with the audit, quality-scores, and gate fields stripped, agreement was measured with Pearson correlation, and the judges were DeepSeek 4.0 and GLM 4.4 in a smoke-test configuration (evidence: `docs/architecture/final-status-2026-08-18.md`, `docs/architecture/innovation-whitepaper-2026-08-18.md`).

2.3 Ten-stage Socratic checklist

Ten checklists, one per stage (topic, protocol, search, screening, extraction, appraisal, analysis, writing, reproducibility, update), each contain ten questions (nine mandatory, one optional). A stage may not begin until the mandatory questions are answered and the checklist gate passes; the gate is enforced by a deterministic script, check_socratic_checklist.py, which has passed full regression. The checklist content draws on the Cochrane Handbook [1], PRISMA 2020 [2], PRISMA-S [3], the MECIR conduct standards, the GRADE Handbook, the living systematic review methods of Elliott et al. [4], and the agreement interpretation bands of Landis and Koch [16] where kappa thresholds are referenced. The purpose of the gate is procedural, not epistemic: it forces the human or agent to state, before entering a stage, what the stage will decide and what evidence would change the decision.

The stage-specific questions are deliberately concrete rather than generic. The topic checklist asks about the nearest existing review and the coverage gap relative to it, about the time-window volume of evidence, and about the falsifiable contribution sentence; the search checklist asks for source-specific strategies, exact dates, exports, and query text; the appraisal checklist asks whether every synthesized result has the required appraisal and whether abstentions were resolved. Because the checklists are JSON files with a fixed schema, they are versioned like code, and the gate script can be regression-tested, which is what allows the project to claim that all ten stages pass the same deterministic gate rather than a sample of them.

2.4 Step-level verifier

Verification in MetaWingman operates at the level of individual methodological steps, not only finished artifacts. The appraisal stage is guarded by verify_appraisal_steps.py, a rule-based verifier covering ten appraisal steps; a step that fails a required check, or that lacks a pending human signature, triggers abstention or opens the human review window, never silent continuation. Appraisal judgments themselves are produced by a six-domain risk-of-bias classification component (domains: selection, detection, attrition, reporting, performance, other), which is described under training below.

The verifier is intentionally rule-based. A learned verifier would itself need verification, and the project' s evidence standard requires that a failed gate be explainable by a deterministic rule, not by a model' s confidence. The ten steps mirror the appraisal dossier structure: each dossier must carry the appraisal tool used, the judgment per domain, the supporting anchor (a quote or table location in the report), the result-level scope of the judgment, and the reviewer signature or its explicit absence. The abstention path matters as much as the pass path: a step that cannot be verified is recorded as abstained, and the human review window is the only route that turns an abstention into a decision. This is the mechanism that produced the abstention-heavy profile observed in the VAL-3 pilot (Section 3.6).

2.5 Audit log and meta-update loop

Every deviation, failure, fix, reflection, question-answer pair, and certificate update is recorded as a JSONL event (event types: deviation, failure, fix, reflection, question_answer, certificate_update) in the audit log. Self-improvement is routed through a meta-update loop: a proposal for change must carry its source, is applied only after passing a human review window, and the applied entry records the commit that implemented it. The loop is the mechanism by which the agent's own failures become training signal for its configuration; at the time of writing this paper, eight audit-log entries had been recorded (Section 3.8).

Two design rules keep the loop from becoming a rubber stamp. First, a proposal without a source is not a proposal: the audit-log schema requires an evidence_source field that names the failing artifact or measurement, and entries without one are rejected. Second, the loop applies to the system's configuration files and documents, not to the review's scientific state; a proposal can change how the next review is conducted, but it cannot silently rewrite an already-frozen protocol. In practice the recorded closures include model-registry corrections after a live API smoke test, a training-weights fix discovered from a crash, and the open-retrieval architecture verdict that replaced a provisional design note. These are small, honest changes, and the paper reports them as such rather than as evidence of general self-improvement.

2.6 Retrieval components

Two retrieval components serve the search and screening stages. The section-role classifier assigns a section role to each passage of a retrieved document so that screening and extraction can target the parts of a report that carry the relevant evidence. The evidence-retrieval component selects candidate passages for extraction. Both are trained models; their measured performance is reported in Section 3.2, and an open-set retrieval comparison that motivated keeping BM25 as the open-retrieval baseline is reported in Section 3.3.

The two components answer different questions and are evaluated differently. Section-role classification is a closed-set labeling task: each passage receives one role, and macro-F1 against the labeled evaluation split is the natural metric. Evidence retrieval is a ranking task within a candidate set: given a review's field-plus-title query, rank the supporting passages, measured by MRR and P@1 on frozen hard-negative pairs. The distinction between candidate-set ranking and open retrieval is not cosmetic; Section 3.3 shows that the trained retrieval model, which ranks well inside a curated candidate set, performs at noise level when asked to retrieve from the full corpus. The design therefore treats the trained components as refiners of a lexical baseline, never as the first-stage retriever.

2.7 Training pipeline

Training data consists of 109,028 examples across two tasks (54,514 per task), split into train (87,264) and dev (21,764); all labels are deterministic weak labels produced by rules, not by models, which makes the labels reproducible by construction (evidence: `docs/architecture/training-run-report-2026-08-17.md`).

The appraisal six-domain component has three generations. The V3 model is trained on rule weak labels and scored on held-out appraisal passages (dev macro-F1 0.8500), a measurement of consistency with the rule labels, not of correctness against a reference standard. A criterion-level evaluation (VAL-2c) scored 100 appraisal criteria against the task' s guideline and found only slight agreement (kappa 0.311, 95% CI 0.191 to 0.431), which we interpreted as a mismatch between the judgment rules and the guideline' s criteria; the component was then re-labeled under criterion supervision, yielding 9,906 re-labeled records (other 8,443, selection 617, reporting 488, performance 145, attrition 112, detection 101). A rubric-based generation (rubric V2) reached dev macro-F1 0.3777 and weighted-F1 0.871, and a generation-2 spot check on 100 items reached agreement 0.80 with weighted-F1 0.871, with kappa not statistically meaningful because 98% of items fell in the majority class (other). Wherever we report these numbers, the claim is consistency with the corresponding labels or rubric, not external validity.

The decision to use deterministic weak labels rather than model-generated labels deserves emphasis. Model-generated labels carry the annotator' s own biases into the training signal, and a model trained on them reproduces those biases with high agreement; the project' s data policy therefore treats any provider-validated JSON object as a candidate label, never as gold. Deterministic rules have the opposite property: they are cheap to audit, versionable, and their failure modes are legible, which is exactly what the VAL-2c evaluation exploited when it found the rules at odds with the guideline. The cost is that rule-consistency ceilings bound what the trained models can reach, and the rubric V2 macro-F1 of 0.3777 is the visible consequence of relabeling against a harder reference: the model now tracks the rubric' s distinctions, but the majority class still dominates the macro average.

2.8 Validation ladder

Validation is organized as a six-level ladder, each level answering a different question:

- VAL-1: select and license-review published-review reconstruction families without contaminating held-out data (first promotion: an analysis-slice case from Hodgkiss et al. [18], BSD-3-Clause materials; remaining families guard-test-only or blocked, recorded per family);
- VAL-2a: encode the AI-only repeated-run plan, the published-expert/corrected-reference policy, integrity exclusions, the error taxonomy, the metric set, and inference limits;
- VAL-2b1: freeze the component-axis task manual —scientific loss weights, release thresholds, configuration and prompt hashes, and stopping rules —before any held-out model run;
- VAL-2b2: fill and freeze reconstruction-case task manuals (first case manual sealed for the analysis slice; remaining families open);
- VAL-2c: freeze a 100-item appraisal-domain spot-check with a sealed an-

swer key and a SHA-256 manifest, and score the weak-label rules against rubric-grounded judgment (kappa 0.311, Section 3.1);
- VAL-3: run the AI-only configuration/ablation pilot (first AI-only screening run reported in Section 3.6; full configuration ladder still open).

Agreement metrics use Cohen' s kappa [17], interpreted with the Landis and Koch bands [16] introduced in Section 2.3; model-vs-model and judge-vs-judge comparisons are reported as Pearson correlations or kappa as stated per experiment. The ladder is deliberately ordered so that the weakest form of evidence (execution) precedes the strongest form currently available (deterministic reconstruction of published numbers); no level above VAL-1 is claimed until the lower levels pass.

Two properties of the ladder matter for how its results should be read. First, each level measures agreement with a different reference, and the references are not interchangeable: agreement with rule labels does not imply agreement with the guideline, which is precisely the failure the VAL-2c result exposed. Second, the ladder does not contain an independent human-reference level for the appraisal component; the criterion-supervised relabeling of VAL-2b2 is a re-anchoring to the guideline, not an independent human judgment. This is a stated limitation of the current evidence, not an oversight, and it is repeated in the Discussion.

2.9 Reconstruction mechanism

Reconstruction follows the living systematic review principle of time slicing [4]: the published review's search cutoff is honored, and only records available before that cutoff are used. For the first reconstruction case, the deterministic analysis pipeline is a locked R pipeline (metafor engine) run three times under identical configuration; the three runs must agree byte-for-byte on outputs before scoring against the published numbers. Scoring uses declared tolerances (pooled effect ±0.05, CI bounds ±0.05, I^2 ±1.0 percentage point, study count exact, Egger significance side). For the ag-rdt living-update family, the search strategies were extracted verbatim from the published supplementary materials, and the later review' s workbook was used as a read-only reference; the operational corpus is a merged, deduplicated slice of PubMed records and preprints assembled under a documented protocol (Section 3.5).

Reconstruction is the project' s answer to the demo problem identified in the Introduction. A capability claim about meta-analysis is hard to falsify on a synthetic example, because the correct answer is whatever the pipeline produced; a published review with published numbers supplies an external anchor. The time-slice rule keeps the exercise honest in the other direction: the reconstruction cannot cheat by using records that were not available to the original team. The same logic extends to the living-update family, where the reference is not a single number but a sequence of included-study sets across versions, and where the scoring tolerances are frozen before the reference is unsealed.

2.10 Ethics and reporting

This study uses only published open-access literature and previously reported data; no human subjects were involved. All experiments were preregistered or protocol-frozen where the ladder requires it (VAL-3 in particular). Reporting follows the spirit of PRISMA 2020 [2] and PRISMA-S [3] for the retrieval components, adapted for a methods paper.

3. Results

All numbers in Section 3 come from machine execution receipts or the project' s result documents; each number' s first occurrence names its evidence source. The receipts and result documents are listed in Data Availability.

3.1 Training corpus and component generations

The frozen training corpus contains 109,028 weakly supervised examples (87,264 train, 21,764 dev), all deterministically labeled (evidence: `docs/architecture/training-run-report-2026-08-17.md`). Table 2 reports the appraisal six-domain component across its three generations. The V3 rule-weak-label model reached dev macro-F1 0.8500 (evidence: `docs/architecture/appraisal-step-component-results-2026-08-18.md`); the rubric V2 generation reached macro-F1 0.3777 and weighted-F1 0.871 (evidence: `docs/architecture/appraisal-rubric-v2-results-2026-08-18.md`); the generation-2 spot check on 100 items achieved agreement 0.80 and weighted-F1 0.871, with kappa uninformative under the 98%-majority-class distribution (evidence: `docs/architecture/appraisal-step-component-results-2026-08-18.md`). The criterion-level evaluation (VAL-2c) scored 100 criteria against the guideline and produced kappa 0.311 (95% CI 0.191 to 0.431), interpreted as a rule-vs-guideline mismatch (evidence: `docs/architecture/val2c-scoring-result-2026-08-18.md`); the resulting criterion-supervised relabeling produced 9,906 records whose class distribution (other 8,443, selection 617, reporting 488, performance 145, attrition 112, detection 101) is heavily skewed toward the majority class, which caps the informativeness of macro-averaged metrics (evidence: `docs/architecture/appraisal-rubric-v2-results-2026-08-18.md`; decision in `docs/architecture/appraisal-task-relabeling-decision-2026-08-18.md`).

Two readings of Table 2 are worth making explicit. The V3 macro-F1 of 0.8500 is high because the model learns the rules that produced its labels, and the rule-labels are internally consistent by construction; that number says nothing about whether the rules match the guideline, which is the question VAL-2c answered with kappa 0.311. After relabeling, the rubric V2 model tracks the harder reference, and its weighted-F1 of 0.871 with a macro-F1 of 0.3777 reflects a task in which 85.2% of records fall in the other class (8,443 of 9,906); a model can be right on the bulk of the data while remaining weak on the rare, decision-relevant domains. The generation-2 spot check (agreement 0.80) is reported with its caveat: with 98% of items in the majority class, agreement is dominated by the majority, and kappa is not a meaningful summary.

3.2 Retrieval components

The section-role classifier reached eval macro-F1 0.9995 on its evaluation split (evidence: `docs/architecture/training-run-report-2026-08-17.md`). The evidence-retrieval component reached candidate-set MRR 0.962 and P@1 0.933 (evidence: `docs/architecture/training-run-report-2026-08-17.md`; receipt-verified in `validation-output/server-download/evidence-retrieval-execution-receipt.json` for loss and recall). These numbers are candidate-set metrics: they describe ranking within a provided candidate set, not open retrieval.

The section-role result at 0.9995 macro-F1 should be read as near-ceiling performance on a closed labeling task with a well-specified label scheme; it does not mean that document structure is understood in any general sense, only that the role of a passage is almost always recoverable from its text and position. The evidence-retrieval numbers likewise describe a narrow job, ranking the correct supporting passage among a small set of hard negatives built for the review's field-plus-title query. Both metrics are reported because they are the ones the components were designed and trained for; the open-set question is handled separately in Section 3.3.

3.3 Open-set retrieval decision

Because the candidate-set numbers do not transfer to open retrieval, we benchmarked single-stage and two-stage retrieval on the full operational corpus (evidence: `docs/architecture/bm25-two-stage-results-2026-08-18.md` and `docs/architecture/retrieval-direction-evidence-2026-08-18.md`). BM25 single-stage (k1=1.5, b=0.75) reached MRR 0.2649, above TF-IDF (0.2199) and far above the trained 110M-parameter model (0.0045), which performed at noise level on open queries. The trained reranker was negative in the two-stage configuration: two-stage MRR fell monotonically with candidate-set size K (0.128 at K=50 to 0.058 at K=200). BM25 recall ceilings were 0.4300, 0.4655, and 0.5071 at K=50, 100, and 200. Two conclusions follow. Open retrieval in MetaWingman uses BM25 single-stage as the baseline, and the trained reranker is reserved for candidate-set refinement where its MRR 0.962 applies. The 0.507 recall ceiling at K=200 also quantifies how much evidence is unreachable by lexical retrieval alone, a bound the search stage must disclose.

The comparison is deliberately blunt. The trained model's open-retrieval MRR of 0.0045 is below random ranking for practical purposes, which rules out using it as a first-stage retriever; and the two-stage decline with K shows that the reranker's errors accumulate as more distractors enter the candidate set, exactly the regime that open retrieval produces. The recall ceiling deserves the same bluntness: if lexical retrieval cannot surface half of the relevant records even at K=200, then a search built on lexical anchors alone is structurally incomplete, and the screening stage must either widen the anchor vocabulary or accept the bound with a documented rationale. These are design constraints measured on one corpus, and the project treats them as corpus-level evidence rather than universal laws.

3.4 First reconstruction case: Hodgkiss et al. 2023

The first time-sliced reconstruction case is the random-effects meta-analysis of peak oxygen uptake (VO2peak) from Hodgkiss et al. [18] (doi:10.1371/journal.pmed.1004082), using the review' s BSD-3-Clause materials and its time slice. The deterministic R pipeline (metafor engine) ran three times under locked configuration; all three runs produced identical outputs, and scoring passed every declared tolerance (evidence: `validation-output/reconstruction-runs/sci-exercise/rep-1/execution-receipt.json` and `docs/architecture/reconstruction-first-case-results-2026-08-18.md`). Table 3 reports the comparison. The pooled mean difference was 2.865 versus the published 2.9 (tolerance ±0.05), the CI was 1.795 to 3.935 versus 1.8 to 3.9, I^2 was 92.67% versus 93% (±1.0 percentage point), the study count matched exactly (k=16), and the Egger test was non-significant on the same side as published (0.483 versus 0.54). The declared boundary matters: this is a recomputation within the same metafor engine family, which demonstrates that our pipeline reproduces published analysis mechanics, not that any model performed the analysis.

Why does this case count as evidence at all if it only re-runs a published analysis? Because the pipeline that produced it is the same pipeline the project uses for the analysis stage, with the same input staging, hash verification, locked environment, and scoring code. A reproduction that passes all five scored dimensions on a real published review, in three byte-identical locked runs, establishes that the mechanical core of the analysis stage is sound: the data staging does not corrupt values, the effect-size computation matches the published engine' s conventions, and the tolerance scheme is calibrated to real published precision. What it does not establish is any claim about model capability, since no model was involved; that is why the case is presented as a pipeline-mechanics result and why the AI-only end-to-end experiment remains the decisive unfinished test.

3.5 Living-update family: ag-rdt (Brümmer et al.)

The second reconstruction family covers the living updates of a systematic review of SARS-CoV-2 antigen rapid diagnostic test (Ag-RDT) accuracy, published by Brümmer et al. in 2021 [19] and updated in 2022 [20]; the 2021 version was subsequently corrected [21]. The 2021 review' s search strategies were extracted verbatim from the published supplementary (S2 Text) with a date-window correction, and the 2022 workbook (CC BY-NC-ND 4.0) was used as a read-only comparison reference (evidence: `docs/architecture/ag-rdt-living-update-case-design-2026-08-18.md`). The operational corpus is a merged slice of 12,498 records: 7,500 PubMed records retrieved via NCBI-native endpoints plus 5,000 preprint records, dedupli-

cated under a documented protocol (evidence: `validation-output/ag-rdt-corpus/merged/execution-rece`
A deterministic screening-chain rehearsal on this corpus showed that lexical term anchors retained 42% of records, which we classify as triage-level retention, and abstract-level extraction coverage was limited (sensitivity 17.4%, specificity 14.7%, n 41.5%) (evidence: `docs/architecture/ag-rdt-living-update-case-design-2026-08-18.md`; receipts in `validation-output/ag-rdt-corpus/screened-rehearsal-v2/execution-receipt.json`). The low extraction coverage quantifies the ceiling of abstract-only processing and motivates full-text retrieval for extraction, a step the pipeline documents as required.

The living-update family tests a different property than the first case: not whether a single pooled estimate can be reproduced, but whether the machinery of versioned evidence, search reuse, and corpus assembly works when the reference moves across versions. The corpus numbers are the measurable part of that test. The 42% retention of the lexical anchors is an order of magnitude above the review' s real inclusion rate, which is exactly the property a triage filter should have: it keeps almost everything that matters while cutting the pool by more than half. The abstract-level extraction coverage (sensitivity 17.4%) is the counterweight: triage at the record level is not extraction at the field level, and the deterministic chain is documented as triage-plus-audit, with field-level extraction deferred to the full-text stage or the AI-assisted arm. The 2022 workbook was used read-only because its CC BY-NC-ND 4.0 license forbids redistribution; the comparison reference is therefore cited, not copied.

3.6 VAL-3: first AI-only screening pilot

The first VAL-3 AI-only screening run used 649 frozen records, of which 149 were gold matches (merged-corpus matches of the 2022 update' s included studies) and 500 were non-gold (evidence: `docs/architecture/val3-ai-screening-pilot-results-2026-08-18.md`). Gold recall was 0.765 (114/149); the dominant loss was abstention (26 of 149 gold records, 17.4%), not false exclusion. The pipeline retained 19.4% of records, versus the deterministic anchors' 42% retention, a substantially more conservative operating point. The pilot ran with a resume-friendly runner over 649 calls with one schema-repair retry before abstention. These numbers support feasibility of AI-first screening with an abstention-heavy profile; they do not establish equivalence with dual independent human screening.

The composition of the recall loss is the pilot' s most informative finding. Twenty-six of the 149 gold records were abstained rather than excluded, which means the loss is recoverable in principle: those records can be routed to the human review window without any re-screening. False exclusion, the error that cannot be recovered without re-running the whole screen, was not the binding constraint. The cost of this design is visible in the retention rate: 19.4% is roughly half the deterministic anchors' 42%, so the AI-first screen keeps a larger candidate pool for the human step. Whether that trade is acceptable depends on the review' s screening volume and the price of missing a record; the pilot' s numbers make the trade explicit rather than hidden in a single accuracy figure.

One pilot on 649 records is a feasibility result, and the preregistered protocol for a full AI-only configuration remains the next step.

3.7 Cross-provider and judge agreement

A cross-provider comparison ran the same blinded set of 999 paragraphs through a GLM model (glm-5.2 C3) and a DeepSeek model (C3-R2) (evidence: `docs/architecture/glm-cross-provider-results-2026-08-18.md`). Section-role macro-F1 was 0.8816 for GLM (999 scored, 0 abstained after three dead-letter recovery rounds; the dominant dead-letter cause was reasoning-token budget exhaustion, all recovered) versus 0.9385 for DeepSeek; retrieval selection accuracy was 0.96 versus 0.93; and across providers, Cohen's kappa on the shared 999 scored passages was 0.8472 (95% CI 0.8221 to 0.8722), interpreted as substantial to almost-perfect agreement [16]. These numbers indicate that the two providers largely agree on section-role assignment, which supports provider diversity for verification roles without claiming that either model is correct where both agree. The certificate double-judge smoke test (DeepSeek 4.0 and GLM 4.4) ran blind with audit, quality-scores, and gate fields stripped and agreement measured by Pearson correlation (evidence: `docs/architecture/innovation-whitepaper-2026-08-18.md`).

Provider diversity is a verification strategy, not a truth oracle. If two independent providers assign the same section role to the same passage, the assignment is more likely to be right than either model alone, but two models can share the same systematic error, and the kappa of 0.8472 does not measure correctness against any reference. The point of the comparison is narrower: the system can run its verification roles on two providers and expect them to agree most of the time, which makes cross-provider double-checking a usable mechanism rather than a source of constant disagreement. The per-metric differences (0.8816 versus 0.9385 on section-role; 0.96 versus 0.93 on retrieval selection) are also a reminder that provider choice moves component performance by a few points, which matters when a component sits at a decision boundary.

3.8 Audit-log closures and system-level evidence

At the time of writing, eight audit-log entries had been recorded, of which seven carried an applied commit through the review window (evidence: `validation-output/audit-log.jsonl`, 8 entries; design in `docs/architecture/innovation-whitepaper-2026-08-18.md`). The ten-stage checklist gate (check_socratic_checklist.py) passed full regression across all ten stages (evidence: `docs/architecture/final-status-2026-08-18.md`). Table 1 maps each of the five components to its mechanism and its evidence.

The eight entries are a process record, not a performance claim. They include corrections to the model registry after a live API smoke test, a training-weight fix, the open-retrieval verdict, the VAL-2c pivot, and the GLM integration lesson, each logged with its evidence source and, where applied, its commit. The honest reading is that the loop works as a bookkeeping mechanism: failures are recorded, proposals carry sources, and applications are traceable. Whether the loop produces durable capability gains is a question that requires longitudinal data the project does not yet have, and the paper does not claim it.

3.9 Comparison with MetaSyn

A task-mapping comparison against MetaSyn [13] covered 15 review-related tasks: 6 fully covered, 7 partially covered, 2 gaps (evidence: `docs/architecture/final-status-2026-08-18.md`; detail in `docs/architecture/metasyn-adapter-desig`
We also audited dataset-card licensing for MetaSyn’ s released assets (MIT-annotated labels with upstream third-party terms) against MetaSyn’ s 422 reviews and 140,585-record corpus (evidence: `docs/architecture/top-journal-contribution-story.md` for the corpus identity; `docs/architecture/metasyn-adapter-design-2026-08-18.md` for the license audit); the audit confirmed the license posture is compatible with research use but that redistribution of third-party text requires upstream terms.

The mapping is a scoping exercise, not a head-to-head benchmark: it compares which tasks each system claims to cover, on the basis of MetaSyn’ s documentation and our own component inventory, without running either system on the other’ s data. The two gaps identified are informative because they point at machinery MetaSyn does not advertise: the gold-corpus evaluation axis and stage attribution of metrics, both of which this paper treats as load-bearing for validation. The license audit matters for a different reason: MetaSyn’ s 422-review and 140,585-record corpus is a natural third-party evaluation axis for retrieval and screening components, and the audit establishes that using it for research evaluation is compatible with the licenses, while redistributing its third-party text is not.

4. Discussion

MetaWingman’ s contribution is architectural rather than a single performance number: a question-first derivation step, verification at the level of individual methodological steps, an auditable self-improvement loop, and reconstruction evidence tied to published reviews. Each of the five components addresses a failure mode that the existing agent literature leaves open. FirstResearch [14] supplies the question-certificate mechanism we migrated; our addition is a double-judge blind evaluation of the certificate and its hard/soft gate structure. MetaSyn [13] is the closest system in scope, an agentic SR/MA pipeline; the task-mapping comparison (6/15 fully covered, 7/15 partial, 2/15 gaps) locates the boundary between the two systems, and our gap analysis identifies where MetaSyn’ s coverage is incomplete relative to the ten-stage contract. Reflexion [10] and process verifiers [11] establish the self-inspection pattern that our step-level verifier and audit loop instantiate in the appraisal domain; our verifier is deterministic and rule-based by design, which trades flexibility for auditability.

The comparison with the wider agent literature is deliberately conservative. Systems such as the AI Scientist [5] and AI Scientist-v2 [6] demonstrate what end-to-end automation looks like when the output is a paper and the review loop is a peer-review proxy; the AI co-scientist [7] shows a generate-debate-evolve loop over hypotheses; Agent Laboratory [8] and PRefLexOR [9] extend the pattern to broader research workflows. MetaWingman does not compete with these systems on coverage. Its claim is narrower and, we argue, more falsifiable: that the review' s scientific contract can be enforced by derivation, step-level verification, and reconstruction, and that each of these mechanisms can be measured. Where the existing systems validate with demos and self-report, this paper validates with published-number reconstruction and execution receipts; where they improve by reflection over finished outputs [10][12], this paper improves by logged proposals that must carry a source and survive a review window. The two styles are complementary, and a production system would plausibly want both.

Three results deserve emphasis because they run against the default assumptions of the field. First, the trained 110M reranker was worse than BM25 on open retrieval (MRR 0.0045 versus 0.2649) and actively harmful in two-stage configuration (0.128 to 0.058 as K grew). Training on candidate sets teaches discrimination within a candidate distribution, and that does not transfer to open queries; the paper' s open-retrieval decision (BM25 baseline, trained reranker only in candidate sets) is a direct consequence of measured data, not of preference. Second, the VAL-2c kappa of 0.311 (95% CI 0.191 to 0.431) between the rule-based appraisal judgments and the guideline' s criteria is the single most informative negative result in the system: it shows that weak-label agreement (macro-F1 0.8500 against rule labels) is not evidence of agreement with the guideline, and it drove the criterion-supervised relabeling of 9,906 records. Third, the reconstruction of Hodgkiss et al. [18] within tolerance on all five scored dimensions, with three byte-identical locked runs, demonstrates that the deterministic analysis chain is reproducible at the level of published numbers; the declared boundary (same metafor engine family) is what keeps this claim honest.

The limitations are substantial and we state them explicitly. (1) All component metrics measure consistency with rule-based or rubric-grounded labels; none of them measures scientific validity against an independent human reference standard beyond the VAL-3 pilot' s gold set. The rubric V2 macro-F1 of 0.3777 alongside weighted-F1 0.871, and the 98%-majority-class distribution of the relabeled data, both limit what macro-averaged scores can tell us. (2) The criterion-level kappa 0.311 documents a rule-vs-guideline mismatch that was repaired by relabeling but not independently re-verified against human judges; the kappa itself has a wide confidence interval (0.191 to 0.431) reflecting evaluation-process variance on a 100-item sample. (3) The reconstruction case is a recomputation within the same statistical engine family; it validates pipeline mechanics, not model capability, and generalizes only as far as metafor-compatible analyses. (4) The VAL-3 pilot is a single AI-only screening run on 649 frozen records with an abstention-heavy profile (gold recall 0.765, retention 19.4%); equivalence with dual independent human screening is not established, and the abstention loss (17.4% of gold) is the main barrier to recall. (5) The abstract-level extraction coverage (sensitivity 17.4%) shows that screening and extraction cannot rely on abstracts, and the full-text retrieval stage required by the pipeline was not part of the numbers reported here. (6) An AI-only end-to-end review, from certificate to published-numbers reconstruction under one protocol, remains unfinished; what we report is a validated assembly of components plus one locked reconstruction slice. (7) Cross-provider agreement (kappa 0.8472) is measured on a single 999-paragraph set with one model per provider; it supports provider diversity as a verification mechanism but says nothing about absolute correctness. (8) The evaluation processes themselves are not yet independently replicated: the VAL-2c kappa, the cross-provider kappa, and the judge correlations each rest on one evaluation run, and the confidence intervals reported for the kappas quantify sampling variance, not replication stability.

Where does the project go next? The immediate target is a complete AI-only end-to-end run on a preregistered case, closing the loop from certificate through screening, extraction, appraisal, and synthesis to a frozen analysis that is scored against the published reference under the VAL-3 protocol. The open problems are the abstention policy (26 of 149 gold records abstained in VAL-3; recall is capped by abstention, not by false exclusion), the appraisal criterion gap (the kappa 0.311 class of failure), and the lexical recall ceiling (0.507 at K=200), which together determine how much of the evidence surface an AI-first pipeline can reach. Two further directions follow from the limitations: an independent human-reference study for the appraisal component, which would give the ladder the level it currently lacks, and a second reconstruction case outside the metafor engine family, which would test whether the pipeline-mechanics result generalizes beyond one statistical engine. Both are preregistration-shaped experiments, and the project’s protocol discipline applies to them as it does to the runs reported here.

5. Conclusion

MetaWingman binds an SR/MA pipeline to a derived, gated question; verifies appraisal at the level of individual steps; records every failure and fix in an auditable log with a source-carrying meta-update loop; and validates its components through a six-level ladder that ends in time-sliced reconstruction of published reviews. The measured record supports three claims: deterministic reproduction of a published meta-analysis within tolerance across three locked runs; component-level consistency with rule- and rubric-grounded labels; and a first AI-only screening pilot with abstention-dominated recall loss. It does not yet support a claim of validated equivalence with human review, and the end-to-end AI-only review remains the decisive unfinished experiment. The design goal of the project is that every number in this paper is reproducible from its execution receipt, which we treat as the minimal standard for methods claims in this field.

Data Availability

All evidence cited in this paper is machine-generated and stored in the project repository under `validation-output/` (execution receipts, JSON) and `docs/architecture/` (result documents, Markdown). Key receipts: the frozen training corpus report (`docs/architecture/training-run-report-2026-08-17.md`), including section-role macro-F1 0.9995 and evidence-retrieval candidate-set MRR 0.962/P@1 0.933, with server receipts in `validation-output/server-download/section-role-executi` and `evidence-retrieval-execution-receipt.json`; appraisal component results (`docs/architecture/appraisal-step-component-results-2026-08-18.md`, `appraisal-rubric-v2-results-2026-08-18.md`, `val2c-scoring-result-2026-08-18.md`, `appraisal-task-relabeling-decision-2026-08-18.md`); open-retrieval benchmarks (`docs/architecture/bm25-two-stage-results-2026-08-18.md`, `retrieval-direction-evidence-2026-08-18.md`); reconstruction receipts (`validation-output/reconstruction-runs/sci-exercise/rep-{1,2,3}/execution-receipt.json` and `docs/architecture/reconstruction-first-case-results-2026-08-18.md`); the ag-rdt corpus and rehearsal receipts (`validation-output/ag-rdt-corpus/**/execution-receipt.json`, `docs/architecture/ag-rdt-living-update-case-design-2026-08-18.md`); the VAL-3 pilot (`docs/architecture/val3-ai-screening-pilot-results-2026-08-18.md`); the cross-provider comparison (`docs/architecture/glm-cross-provider-results-2026-08-18.md`); the certificate double-judge smoke test and the audit-log design (`docs/architecture/innovation-whitepaper` `docs/architecture/final-status-2026-08-18.md`); the audit-log closures (`validation-output/audit-log.jsonl`); and the MetaSyn task mapping, corpus identity, and license audit (`docs/architecture/final-status-2026-08-18.md`, `docs/architecture/top-journal-contribution-story.md`, `docs/architecture/metasyn-adapter-design` Analysis code and the deterministic R reconstruction pipeline are versioned in the repository; the reconstruction inputs use the published review's BSD-3-Clause materials, and the 2022 ag-rdt workbook is used read-only under its CC BY-NC-ND 4.0 license. Raw PDF/XML corpora are kept outside Git per the project's data policy.

Ethics Declaration

This study did not involve human subjects, patient data, or animal experiments. It uses published open-access literature, publicly released benchmark materials, and previously reported data. All reuse respects the source licenses (BSD-3-Clause for the Hodgkiss materials; CC BY-NC-ND 4.0 read-only for the 2022 ag-rdt workbook; MIT annotations with upstream third-party terms for the MetaSyn assets).

Author Contributions (CRediT)

[Placeholder author list; to be completed by the author team.] Conceptualization: [Author A]; Methodology: [Author A], [Author B]; Software: [Author A], [Author B]; Validation: [Author A], [Author B]; Formal analysis: [Author B]; Data curation: [Author B]; Writing (original draft): [Author A]; Writing (review and editing): [Author A], [Author B]; Project administration: [Author A].

Competing Interests

The authors declare no competing interests.

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AI-use Statement

This manuscript was drafted with AI assistance and verified against the project's evidence dossiers and execution receipts. All reported numbers originate from machine execution receipts or versioned result documents generated by the MetaWingman pipeline; no number was generated during manuscript writing. The draft follows a prespecified evidence dossier, and each figure was checked against its cited receipt. The ten-stage checklist gate and audit-log discipline of the project were applied to the manuscript itself.

Tables

Table 1. The five components of MetaWingman, their mechanisms, and the evidence reported in this paper.

Component	Mechanism	Evidence reported
Review Question Certificate	Seven-stage derivation (primitives to falsifiable hypothesis), hard novelty gate (Europe PMC) and soft answerability gate, failure-update rule; max_tokens 8192	Double-judge blind evaluation (DeepSeek 4.0, GLM 4.4), Pearson agreement; hard/soft gate structure
Ten-stage Socratic checklist	Ten stages × ten questions (9 mandatory + 1 optional); deterministic gate script	Full regression pass of check_socratic_checklist.py; sources: Cochrane Handbook, PRISMA 2020, PRISMA-S, MECIR, GRADE Handbook, Elliott et al.

Component	Mechanism	Evidence reported
Step-level verifier	verify_appraisal_steps.py: ten appraisal steps, abstention and human-review window	Six-domain RoB classification component (Table 2)
Audit log + meta-update loop	JSONL events (deviation/failure/fix/reflection/question_answer/certificate_update); source-carrying proposals applied via review window with commit recorded	Eight audit-log entries recorded, seven applied (Section 3.8)
Retrieval components	Section-role classification; evidence retrieval	Section-role macro-F1 0.9995; retrieval MRR 0.962, P@1 0.933

Table 2. Appraisal six-domain component across generations. All labels are deterministic weak labels unless stated; metrics measure consistency with the stated reference, not external validity. Sources: `docs/architecture/appraisal-step-component-results-2026-08-18.md`, `appraisal-rubric-v2-results-2026-08-18.md`, `val2c-scoring-result-2026-08-18.md`.

Generation / evaluation	Reference	Metric	Value
V3 (rule weak labels)	Rule labels on dev	macro-F1	0.8500
Rubric V2	Rubric-grounded labels on dev	macro-F1 / weighted-F1	0.3777 / 0.871
Generation-2 spot check (n=100)	Rubric-grounded labels	agreement / weighted-F1	0.80 / 0.871
VAL-2c (n=100 criteria)	Guideline criteria	Cohen’s kappa (95% CI)	0.311 (0.191 to 0.431)
Criterion-supervised relabeling	Guideline criteria	Records (other/selection/reporting/performance/attrition/detection)	9,906 (8,443/617/488/145/112/101)

Table 3. Reconstruction of Hodgkiss et al. 2023 [18]: deterministic R pipeline versus published values. Three locked runs produced byte-identical outputs (evidence: `validation-output/reconstruction-runs/sci-exercise/rep-1/execution-receipt.j`

Dimension	Reconstructed	Published	Tolerance	Verdict
Pooled MD (mL/kg/min)	2.865	2.9	±0.05	Pass (Δ −0.035)
CI lower	1.795	1.8	±0.05	Pass (Δ −0.005)
CI upper	3.935	3.9	±0.05	Pass
I^2	92.67%	93%	±1.0 pp	Pass (Δ −0.33)
Studies k	16	16	Exact	Pass
Egger test p	0.483	0.54	Significance side	Pass (both non-significant)

Table 4. Cross-provider comparison on the same blinded set of 999 paragraphs (evidence: `docs/architecture/glm-cross-provider-results-2026-08-18.md`).

Metric	DeepSeek C3-R2	GLM glm-5.2 C3	Agreement
Section-role macro-F1	0.9385	0.8816	—
Retrieval selection accuracy	0.93	0.96	—
Cohen’s kappa on shared 999 scored passages	—	—	0.8472 (95% CI 0.8221 to 0.8722)

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